

Contents

I	Parameters of Leavitt Law	2
II	Galactic Cepheid Dataset and Calibration Method	3
1	Galactic Cepheid Dataset	4
1.1	Dataset Cleaning	5
1.1.1	Pulsating in Overtone	5
1.1.2	Photometric error	5
1.1.3	PL/PW Residuals	7
1.2	Extinction Law Calibration	7
1.3	Cleaned Dataset	9
1.4	Leavitt Law	11
2	Calibration Method	13
2.1	Leavitt Law residuals and corresponding Wesenheits	13
2.2	Residual Analysis	15
2.3	Decoupling $\delta\mu - \delta E_{BV}$	18
2.4	Calibrated Leavitt Law	20
III	Result Implications	23
3	Discussion	24
3.1	Validation of Result : Cluster Cepheid	24
3.2	Reddening Comparison	24
3.3	Distance Comparison	25
3.4	Luminosity Comparison	25
4	Conclusion	26
4.1	Result	26
4.2	Application Of The Result	26
4.2.1	Understanding the Universe: 1990s	26

4.2.2	Cosmic Distance Ladder and the Hubble Constant: The Twenty- First Century	29
4.2.3	Instruments of furture	29
	Bibliography	30
	Bibliography	31

Part I

Parameters of Leavitt Law

Part II

Galactic Cepheid Dataset and Calibration Method

1 Galactic Cepheid Dataset

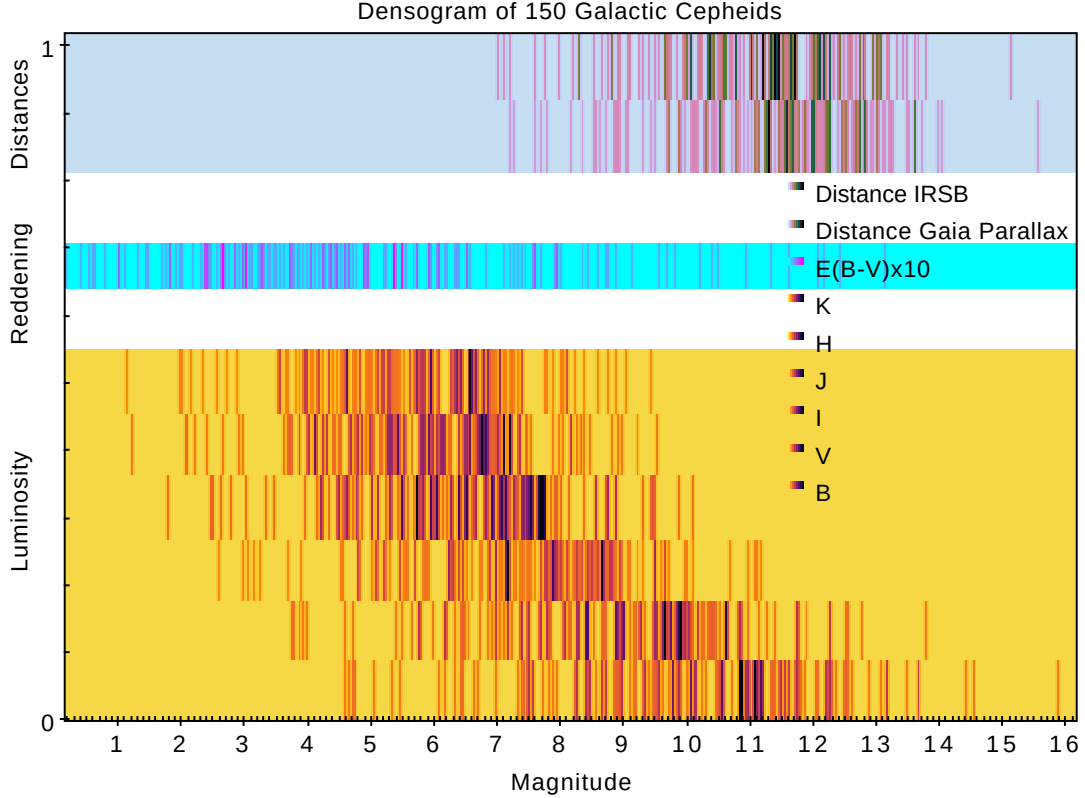


Figure 1.1: Densogram showing the distribution of 150 Galactic classical Cepheids. The top two rows present distance distributions derived from two independent methods, highlighting subtle differences between the results. The third row shows reddening estimates, scaled by a factor of ten for visualization. The remaining row display photometric measurements in the K , H , J , I , V , and B bands, ordered from top to bottom.

A multi-band photometric dataset of 150 Galactic Classical Cepheids (GCCs) is provided by Dr. Jesper Storm to me for the calibration of distance and reddening estimates. The dataset includes apparent magnitudes in the B , V , R , I , J , H , and K bands, obtained using different instruments at multiple observing sites (SGF⁺11). For each Cepheid, color excess $E(B - V)$ is estimated following the method of (Fer94). Distance moduli are derived using three independent techniques: the infrared surface brightness (IRSB) method (SGF⁺11), Gaia parallaxes for the full sample, and cluster membership distances for 17 Cepheids associated with stellar clusters (CA23). The pulsation period is measured in days and transformed in logarithmic scale (not shown in the plot.).

1.1 Dataset Cleaning

The prepared dataset is collected from different sources, therefore some Cepheids could be inconsistent in certain ways - pulsation mode, photometric error, etc., for data calibration. Such stars considered as outliers and must be filtered out is the first step.

1.1.1 Pulsating in Overtone

In the given dataset, seven Cepheid found to be overtone, and have excluded from the dataset. Those are namely: **SU Cas**, **V0465 Mon**, **QZ Nor**, **DT Cyg**, **EV Sct**, **FF Aql** and **SZ Tau**. The dataset of 143 Galactic Classical Cepheids, pulsating in fundamental mode, is given in Appendix A of this thesis.

1.1.2 Photometric error

Cepheids with inconsistent photometric measurements must be removed from the dataset. To identify the outliers, the period-color relation is employed. This approach is effective because both the intrinsic color and the period of a Cepheid are independent of distance. Since Cepheid's period strongly correlates with its luminosity, plotting the period against color mimics the features of the color-magnitude diagram — particularly representing the region known as the instability strip in color magnitude diagram.

Within the instability strip, short-period Cepheids appear lower, while long-period (and thus brighter Cepheids) appear higher. Cepheids that significantly deviate from the linear trend of the instability strip are likely affected by incorrect photometry in specific bands and are therefore considered as outliers. In general, the intrinsic color index for any pair of photometric bands can be calculated as shown in equation 1.1.

$$(m_1 - m_2)_0 = (m_1 - m_2) - (R_1^{BV} - R_2^{BV})E_{BV} \quad (1.1)$$

From BVIJHK photometry fifteen color combinations are made. Tracing the anomalous movement of individual GCCs in each color-period plot suggests inconsistency in photometric measurement in particular bands. Cepheids with such anomalous behavior are listed in the Table 1.1

Table 1.1: List of 13 photometric outliers (out of 143 GCCs) with corresponding color index

Index	Cepheid	Color index (marking outlier)
62	RU Sct	J-H, J-K
67	SU Cru	B-K, V-H, V-K, I-H, I-K, J-H, J-K
68	KQ Sco	V-J
71	V496 Aql	B-I, V-I, I-J, I-H, I-K,
126	TW Nor	V-J , V-K
127	V0508 Mon	B-V
128	TZ Mon	B-V
129	AA Ser	B-I, B-J, B-H, B-K, V-I, V-J, V-H, V-K, J-H,
130	CN Sct	B-I, B-J, B-H, B-K, V-I, V-J, V-K, J-H,
135	HZ Per	B-I, B-J, B-H, B-K, V-I, V-J, V-H, V-K, I-J, J-H,
136	CT Car	B-V, B-I, B-J, B-H, B-K, V-I, I-J, I-H, I-K,
137	TY Mon	B-V, B-I, B-J, B-H, B-K, V-I, I-J, I-H, I-K,
138	AA Mon	B-V, B-I, B-J, B-H, B-K

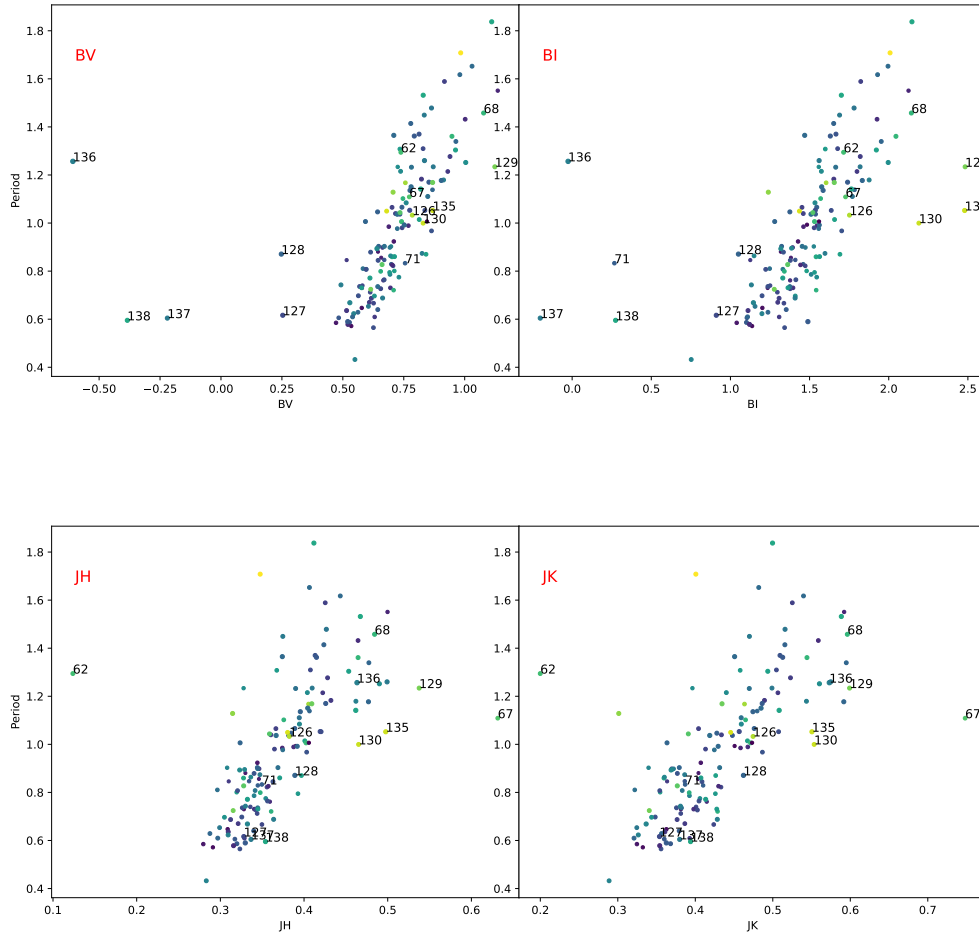


Figure 1.2: Outliers are filtered using 15 combinations of period–color plots. Out of 143 Cepheids, 13 GCCs arbitrarily move away from the generic trend of the instability strip for certain color combinations. These sources are marked with their respective index numbers and are listed in Table 1.1. Remains 130 Cepheids.

Upon removing photometric outliers from the dataset of 143 Cepheids, 130 Cepheids remain available for further analysis.

1.1.3 PL/PW Residuals

Another set of outliers selected from calibration process. To calibrate individual GCC for distance and reddening error, respective residuals from PL and PW relations are correlated. The slope of the regression line of such correlation plays a central role in the calibration process. In this process, nine Cepheids (**CY Aur, RX Cam, V1344 Aql, RW Cam, SV Per, MW Cyg, RZ Gem, V0495 Cyg and VX Per**), found be of large residue, which significantly affecting the slope of regression line as compare to remaining dataset, therefore rejected from the analysis. Ultimately leaving 121 Cepheids in the cleaned dataset.

1.2 Extinction Law Calibration

To calibrate the reddening-distance error, pulsation period of Cepheid plays the key role which is a well observed parameter for individual star. For GCCs, period indicates the true absolute magnitude over the spectral range, which aids to decouple distance-extinction from apparent magnitude. Unlike distance, extinction cannot be directly observed however deviation of extinction across bands can be measured as color excess. For my dataset, color excess is measured between B band and V band, i.e. $E(B - V) = A_B - A_V$. By scaling with reddening ratio, color excess can be converted into extinction i.e. for visual band $R_V \times E(B - V) = A_V$. Here reddening ratio is related with interstellar grain size increases with particle size, for Milky Way it is expected to be $R_V = 3.23$ (STR04). For other bands, composite reddening ratio can be derived using an extinction law, $R_V \times \frac{A_\lambda}{A_V} = R_\lambda$ (CCM89). In his calibration method, Madore (MFM17) adopted the extinction law given in table 4 of (FAS⁺07) to derive (V-I) based wesenheit magnitude of 59 GCCs.

Wesenheit function, $W = m - R(m_1 - m_2)$, is sensitive to composite reddening ratio, implies error in adopted extinction law as well as visual reddening ratio would contribute to residue of period-wesenheit relation. Therefore, it is crucial to minimize such systematics prior to distance-reddening calibration of given GCCs dataset.

Since wesenheit magnitude do not require measurement of color excess and yields the same result as of unreddened magnitude, $W = W_0$, leveraging over available color excess measurements, deviation between W and W_0 can be plotted for the given dataset.

Noticing the non-zero deviation, it can be said that the deviation, δW , can be due to following equation:

$$\delta W = \delta A_\lambda - \delta R \times E(B - V)$$

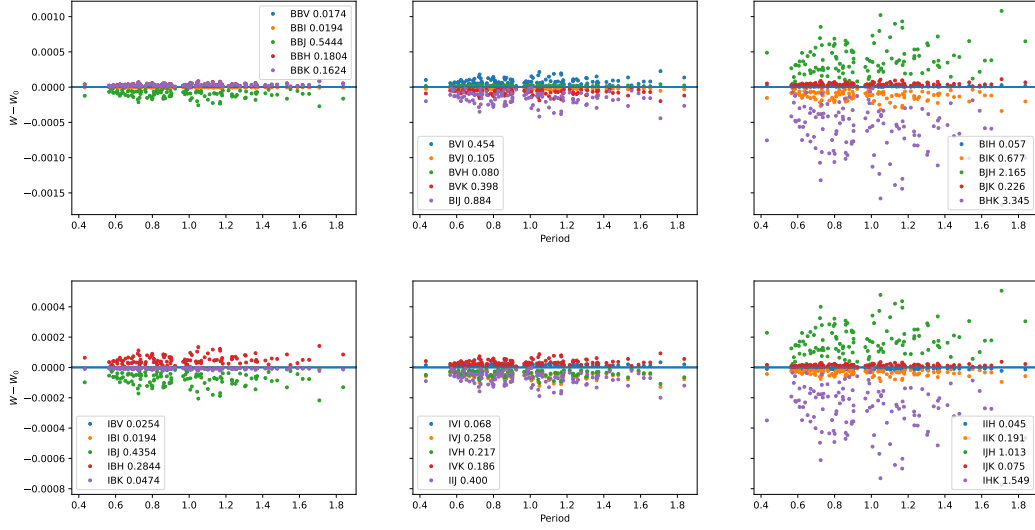


Figure 1.3: Deviation between reddened and unreddened weseneit magnitude. (FAS⁺07) Extinction law and Reddening ratio (STR04) $R_V = 3.23$

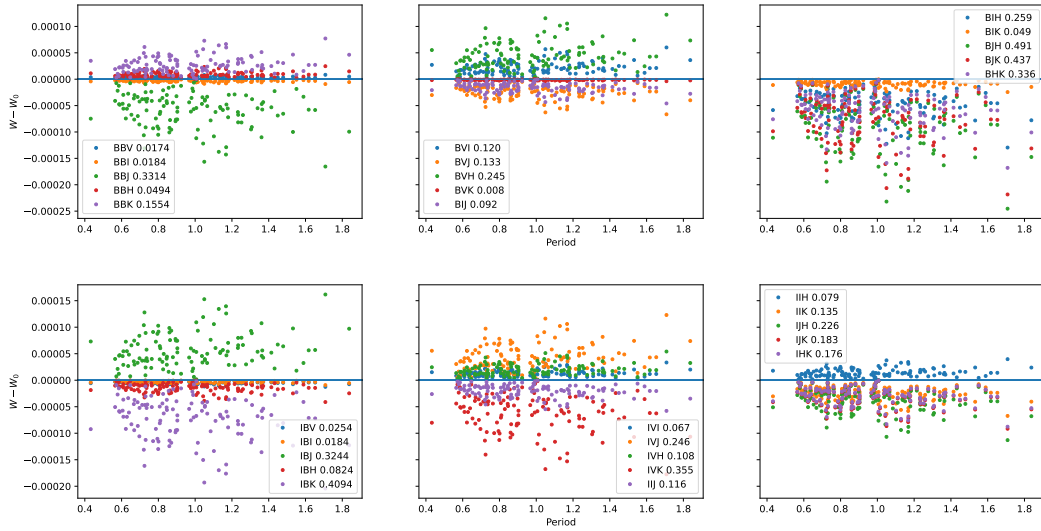


Figure 1.4: Deviation between weseneit magnitude with ajusted extinction law and reddening ratio,

rsde

For given dataset, I adjusted the reddening ratio and extinction ratio to different values, see Table 1.2, for which the deviation between W and W_0 approaches to the least.

Table 1.2: Extinction Law A_λ/A_V comparison with updated value with corresponding reddening ratios

Bands	Fouque (2004) with $R_V = 3.23$	Adjusted with $R_V = 3.19$
B	1.31	1.31
V	1.0	1.0
I	0.608	0.608
J	0.292	0.2969
H	0.181	0.1805
K	0.119	0.1231

Such fine adjustment in reddening ratio and extinction law for considered dataset yields smaller deviation between W and W_0 , minimizing the systematic in residue of period-wesenheit magnitude.

1.3 Cleaned Dataset

After cleaning the outliers, the dataset can be used for the calibration of Leavitt Law. Following step would transform observed apparent magnitude into true apparent magnitude using extinction law (FAS⁺07) and color excess measurements. This is shown in the adjacent plot - red region transforming into blue region. In the plot, period (x-axis) plotted against luminosity (y-axis). The apparent magnitude, B and K magnitude depicted with red dashed line. Remaining bands - V, I, J and H lie in between the two edges of colored area. After band wise extinction correction, true apparent magnitude results as depicted by blue region. Note that the shift in B band is much larger than K band, as light of short wavelength is much more sensitive to extinction compare to longer ones. After extinction correction, resulting magnitude becomes brighter and the area of red region squeezed to blue region. At this point, no clear correlation between the period and luminosity is evident.

$$m_0 = m_\lambda - R_\lambda * E_{BV} \quad (1.2)$$

Following step adjusts the distance for each Cepheid, transforming true apparent magnitudes into true absolute magnitudes, as depicted by the yellow region. At this point, a linear trend begins to emerge between the period and intrinsic luminosity of the Cepheids which known as the Leavitt Law.

$$M_0 = m_\lambda - R_\lambda * E_{BV} - \mu \quad (1.3)$$

Another version of extinction free luminosity (true absolute magnitude) is reddening free magnitude which derived through Wesenheit function (Mad82). An important aspect of the Wesenheit magnitude is that does not require the measurement of color excess, rather depends on color index. For multiband photometry (BVIJHK), I considered all the fifteen permutations of color indices to understand the key differences between them. It is crucial because wesenheim magnitude for respective band m_λ is derived from reddening ratio $R12_\lambda$.

For residual correlation of period-luminosity relation and period-wesenheit relation Composite Leavitt Law (CLL) them for the Leavitt Law calibration. The cyan and green regions in the plot represent two versions of the Wesenheit magnitudes—HK and JH, respectively. The lower and upper edges of these regions correspond to the B and K bands, while the shaded region encompasses the remaining four bands used in the Wesenheit magnitude. All other versions of the Wesenheit magnitude lie between the HK and JH regions, as explicitly shown in Figure ??.

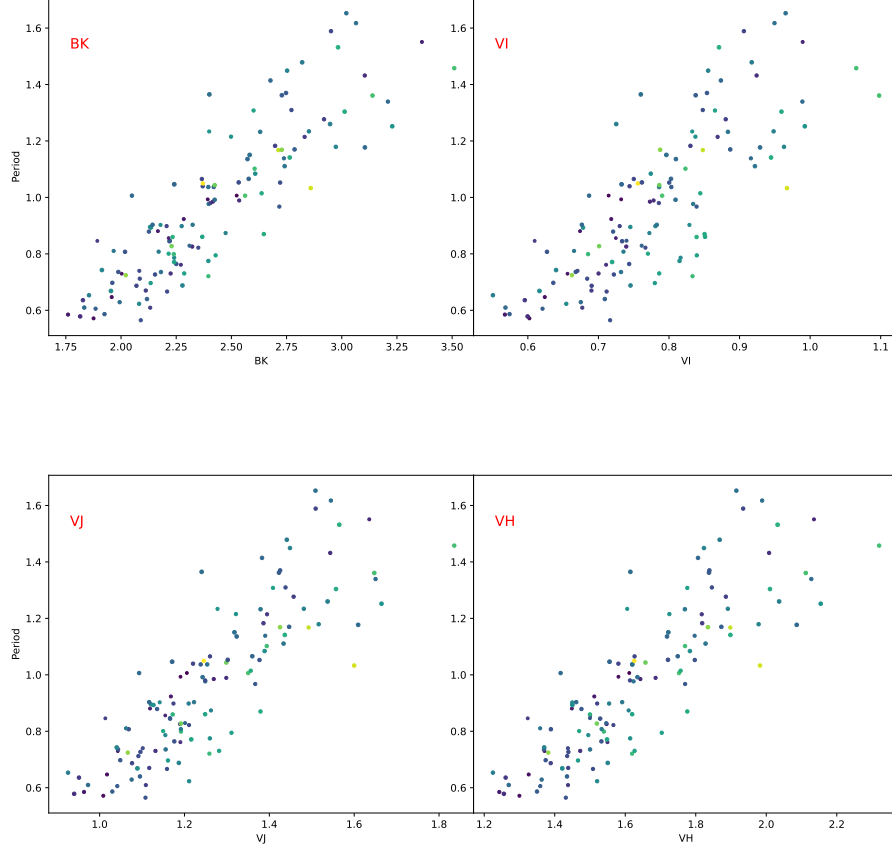


Figure 1.5: Out of 125 GCCs, 4 stars influencing the slope of residual regression line significantly, consequently affecting reddening-distance error estimation for all stars. They are also discarded therefore 121 Cepheids are

The four subplots of Figure ?? compare all fifteen versions of the Wesenheit magnitudes. Each plot includes the HK and JH Wesenheit magnitudes in the background as a reference for ease of comparison. It is clear that not all color bases for the Wesenheit lead to the same magnitude. For example, the area covered by the B-to-K Wesenheit of HK is much larger than that of the BV-based or other Wesenheit magnitudes. Lesser difference between the two edges of a given Wesenheit makes it a better choice for the calibration process, as all Wesenheit magnitudes of that color suggest the same reddening-free magnitude and only contain errors due to the distance modulus. Based on this observation, the VI, VK, IH, and JK-based Wesenheit magnitudes are selected for the analysis.

1.4 Leavitt Law

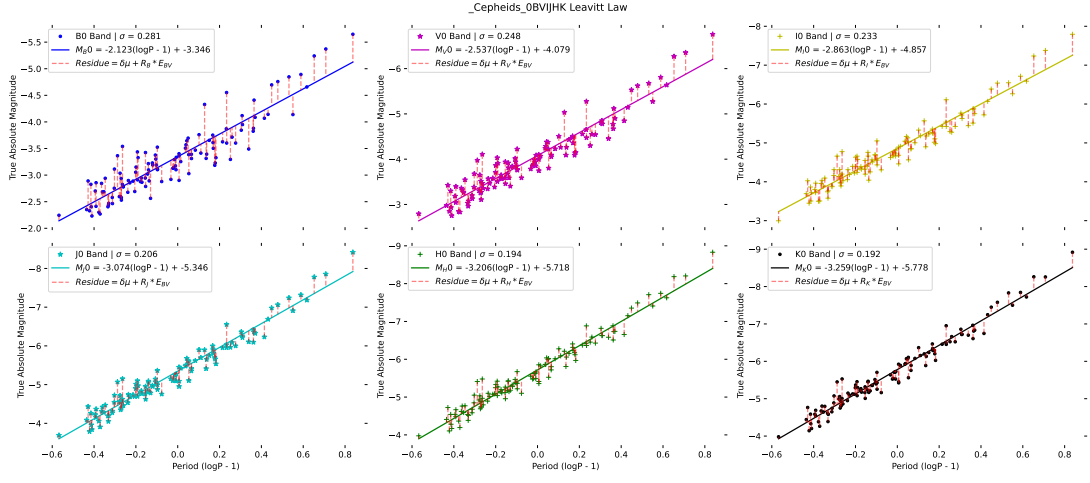


Figure 1.6: Leavitt Law based on 121 Galactic Cepheids with Gaia distance prior to calibration

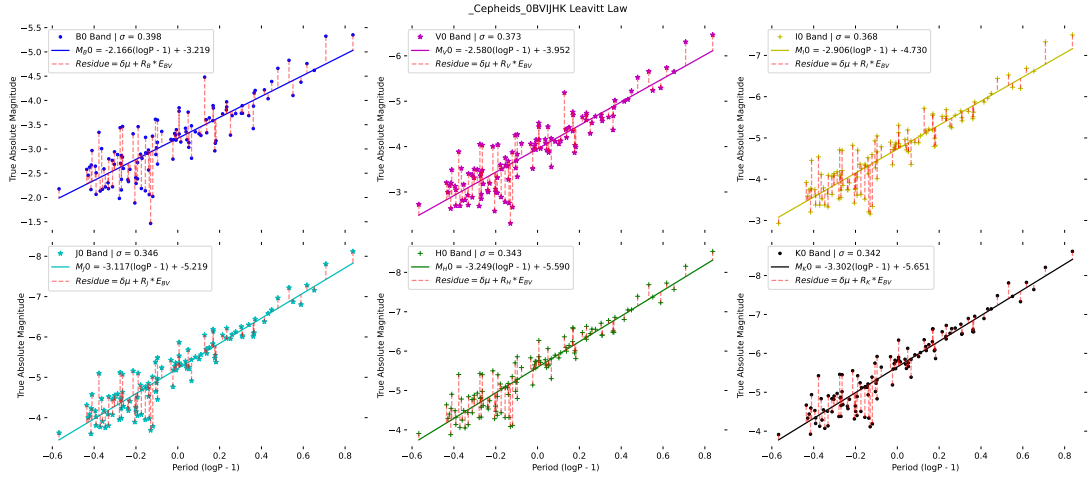


Figure 1.7: Leavitt Law with IRSB distance prior to calibration

2 Calibration Method

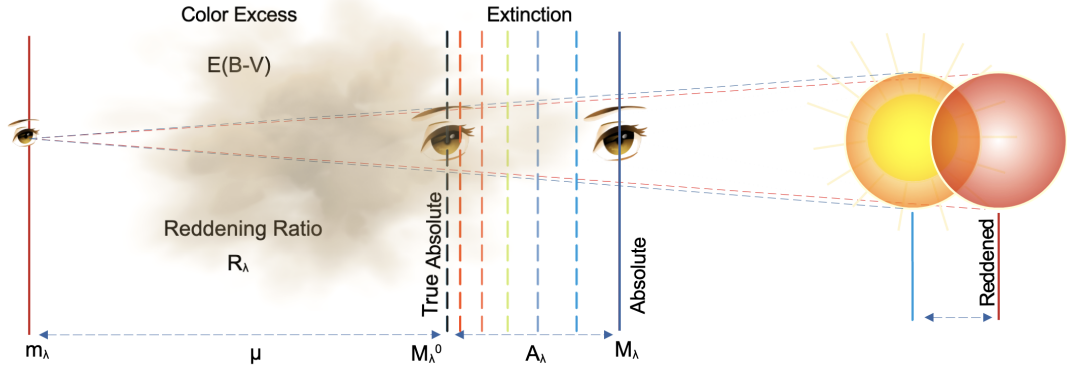


Figure 2.1: The Senario

2.1 Leavitt Law residuals and corresponding Wesenheits

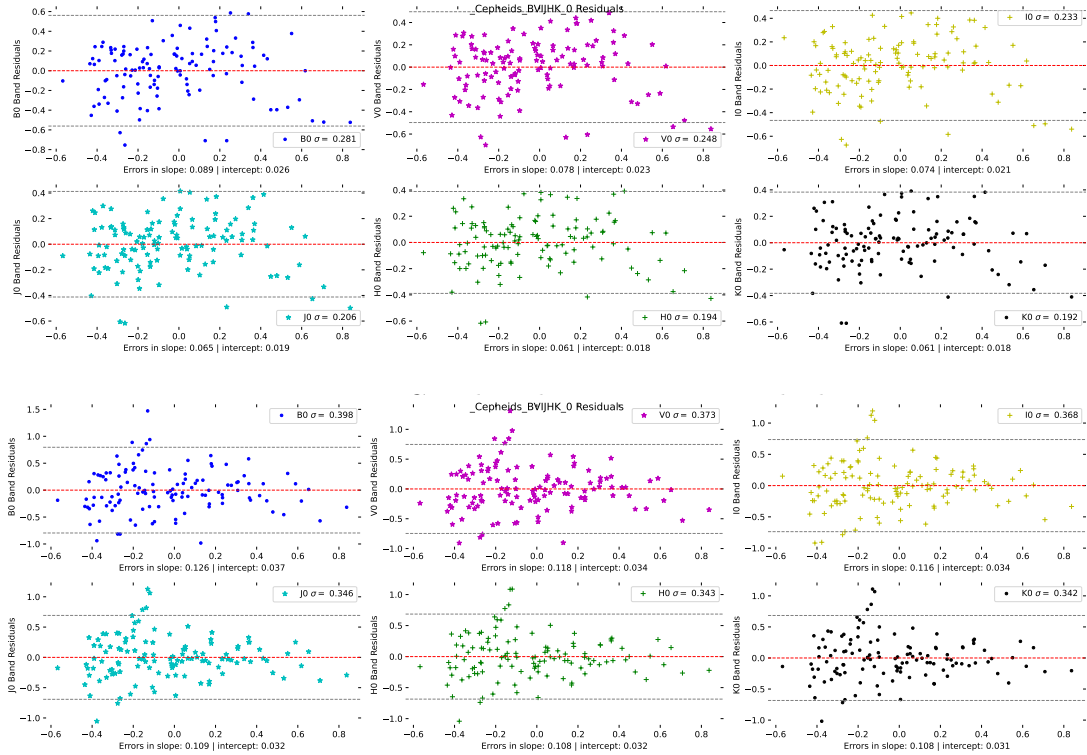


Figure 2.3: IRSB Leavitt Law Residual

Observations in different bands of light provide information about the effect of inter-

stellar reddening on light. For longer wavelengths, the impact of reddening decreases significantly, implying a reduced possibility of reddening errors and leading to a tighter Leavitt Law. In its most generic form, the linear fit to the dataset is expressed as follows:

$$M_\lambda = \alpha_\lambda(\log P - 1) + \beta_\lambda \quad (2.1)$$

$$W_\lambda^{12} = \alpha_\lambda^{12}(\log P - 1) + \beta_\lambda^{12} \quad (2.2)$$

Since the periodic range of Cepheids lies between 0.49 and 1.8 on a logarithmic scale, 1 is chosen as the pivot point and subtracted from the period before calculating the regression coefficients, i.e., the slope and intercept. The Leavitt Law for each band is shown in Figure 2.14, including four selected Wesenheit variants for VI, VK, IH, and

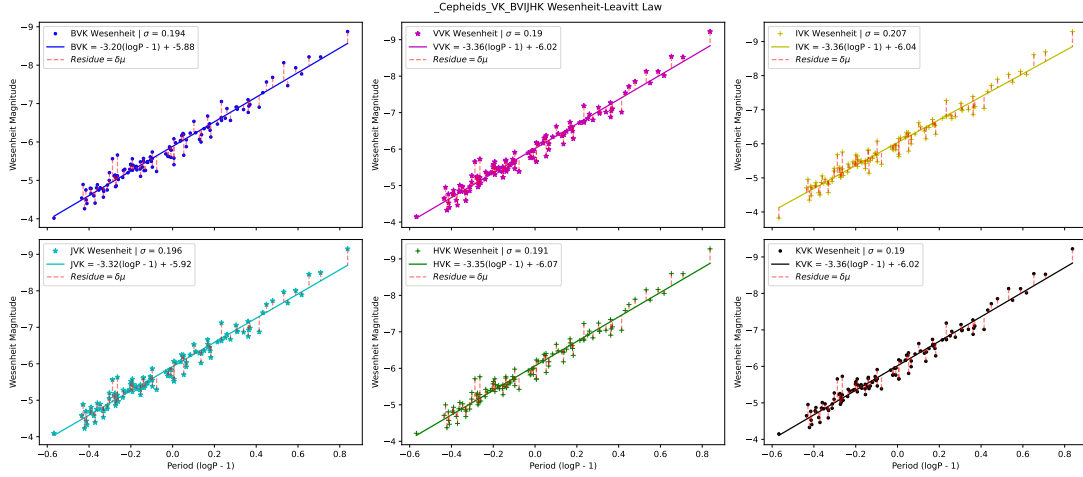


Figure 2.4: Gaia Leavitt-Wesenheit Residual

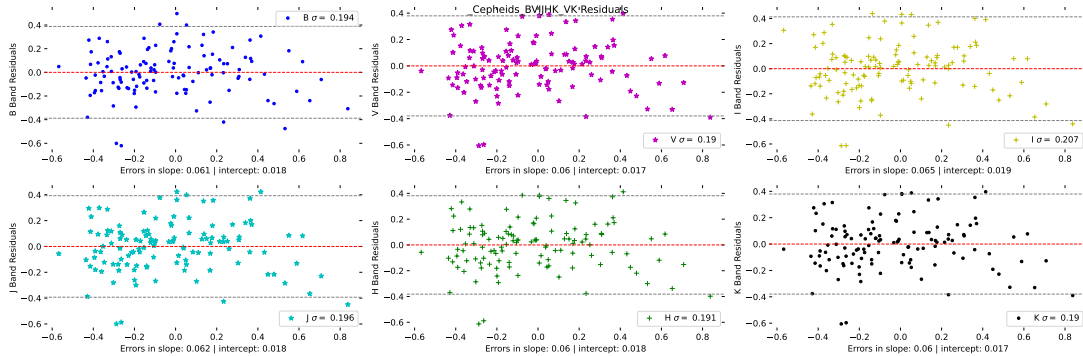


Figure 2.5: Gaia based Leavitt Law Residuals

The plots above provide a comparative overview of the Leavitt Laws across different bands. In the first column, notice the decreasing width of the PL relation from the B-band to the K-band. The increasing Pearson correlation coefficient indicates a reduction

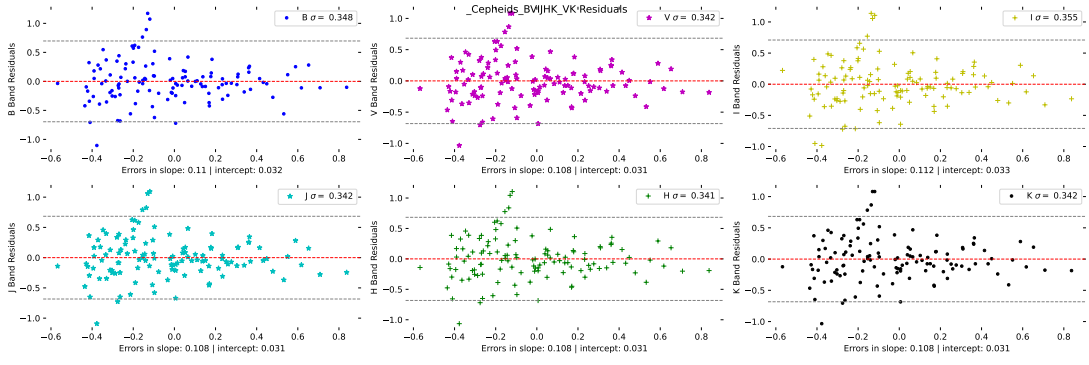


Figure 2.6: IRSB Leavitt Law Residual

in residuals. Assuming Cepheids align perfectly with the PL relation suggests that the residuals must stem from errors in reddening or distance measurements.

The contribution of residuals due to extinction errors varies with the band, while errors from the distance modulus remain constant. Therefore, the decrease in the width of the PL relation is strongly related to incorrect reddening measurements.

`clean/PLW_mc.pdf`

Figure 2.7: Slope and Intercept of BVIJHK PL and PW relations. Interestingly, PW seems constant for all the bands, which is not the case for PL relations.

The four variants of the PW relation have relatively higher r-values (approaching -1) than their

corresponding PL relations, as Wesenheits are independent of extinction-based errors. Being reddening-free, the scatter in the Wesenheit-based Leavitt Law primarily arises from errors in distance and the reddening ratio. In this research, the reddening ratio is assumed to be correctly known ($R_V = 3.23$, (STR04)), making distance the only source of the scatter.

The slope and intercept of each PL and PW relations are summarized in Table ?? and Figure 2.7. The Table ?? also list the uncertainty in slope and intercept of PL and PW relation, arising due to a larger scatter. After the calibration, these uncertainty will be deminished significantly.

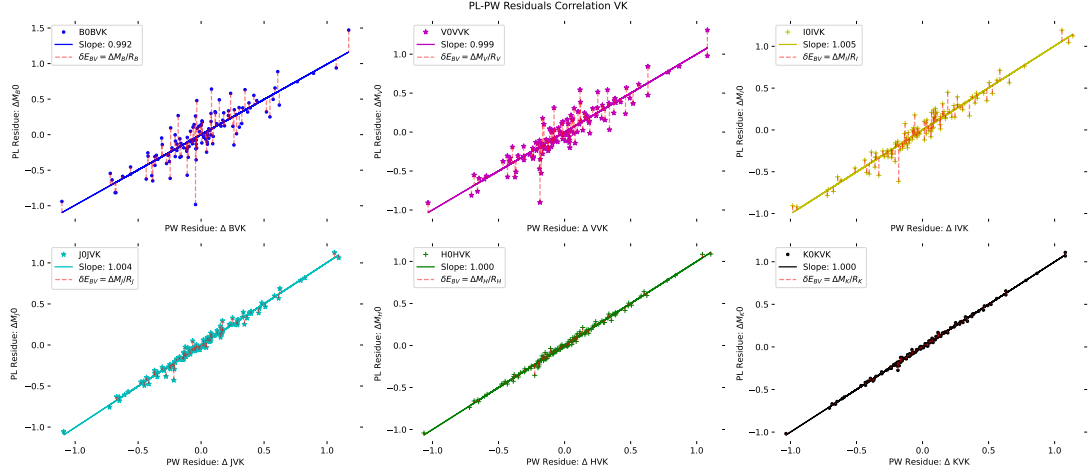
On the other hand, note the convergence of slope and intercept of PW relations at K band for VI, VK and IH in the adjacent figure. JK based PW relation showing a fine deviation from the rest. These subtle variation in PW slope and intercept suggests a fine-tuning of reddening law for longer wavelength, though such exercise is out of the scope of this thesis. In the next section, procedure for analysing these residual of PL and PW relations is briefly discussed.

2.2 Residual Analysis

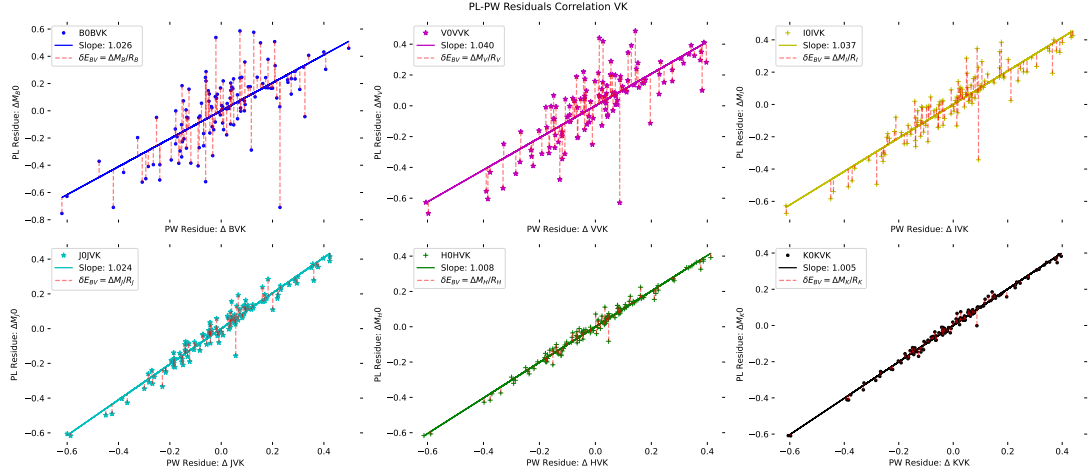
The residuals of PL and PW relations are calculated as follows:

$$\Delta M_\lambda = M_\lambda - (\alpha_\lambda \log P + \beta_\lambda) \quad (2.3)$$

$$\Delta W_\lambda^{12} = W_\lambda^{12} - (\alpha_\lambda^{12} \log P + \beta_\lambda^{12}) \quad (2.4)$$



(a) IRSB $\Delta W_\lambda^{VK} - \Delta M_\lambda$ correlation



(b) Gaia $\Delta W_\lambda^{VK} - \Delta M_\lambda$ correlation

Figure 2.8: $\Delta W - \Delta M$ correlation plots shown for VK wesenheit - IRSB and Gaia. With increasing wavelength, slope is approaching to 1.

Here, ΔM_λ represents the residuals of the PL relation, and ΔW_λ^{12} represents the residuals of the PW relation. Since distance modulus μ term is involved in both the parameters, M_λ and W_λ , thereby $\delta\mu$, impacts both ΔW and ΔM equally. However, δE_{BV} does not contribute to ΔW , and only affects ΔM by an amount of $\delta E_{BV}/R_\lambda = \delta A_\lambda$. Since other possible factor like the effect of metallicity is not considered in this work, the equation for PL and PW residuals, equations ?? and ??, can be expressed as:

$$\Delta M_\lambda = -(R_\lambda^{BV} * \delta E(B - V) + \delta\mu)$$

$$\Delta W_\lambda^{BV} = -\delta\mu$$

Geometrically, these equations suggest that δE_{BV} (scaled by R_{λ}^{-1}) shifts a star along the y-axis (vertically), while $\delta\mu$ affects both the x and y axes equally, moving the star along a line with slope 1. For the complete ensemble of stars, the slope of the residual correlation, ρ , when deviating from 1, suggests a significant contribution from reddening errors, as clearly observed for the case of the B and V bands in Figure 2.8. For the longer wavelengths—J, H, and K—the contribution from reddening errors is minimized, and distance-based errors become dominant, causing the slope ρ to approach 1.

Tracing an individual Cepheid in every $\Delta W_{\lambda}^{12} - \Delta M_{\lambda}$ correlation plots provides valuable insight into the extinction error associated with each band. By converting the extinction error into reddening error for each band and averaging these val-

Figure 2.9: For each star, multiple trials are conducted to estimate the reddening-distance error pair in the B band. This process is then repeated for the remaining bands to constrain the correct $\delta\mu - \delta E_{BV}$ error pair. (MFM17)

ues, an estimate of the reddening correction for that specific star is obtained. However, at this stage, the correction for the modulus is not yet determined. To estimate the modulus error, a series of trials for $\delta\mu$ are conducted, which shifts the star along the line of slope 1, as shown in Figure 2.9 for stars B and C. The solid black diagonal line represents the regression line derived from one of the $\Delta W - \Delta M$ plots. The vertical deviation from this regression line, for each trial, reflects the amount of reddening correction needed. Several trial combinations for star B are labeled as 'B0' and 'B1', while 'C2', 'C3', and 'C4' correspond to the trial positions for star C.

These examples illustrate multiple possible error pairs for the B-band. Similar correlation plots are also constructed for the VIJHK bands. Each residual correlation plot provides the corresponding reddening errors for the respective $\delta\mu$ trials, i.e. every $\delta\mu$ trial corresponds to six δE_{BV} . The $\delta\mu$ trial that minimizes the variance of δE_{BV} across the bands identifies the correct distance correction trial. This ultimately decouples the error budget between $\delta\mu$ and δE_{BV} for each star.

This method of decoupling errors, introduced by (MFM17), involves correlating ΔW_V^{VI} with ΔM_{λ} . In contrast, I have used the composite Wesenheit function, ΔW_{λ}^{VI} , for each correlation instance. The key motivation behind this choice is the definition of the Wesenheit function, which provides a reddening-free magnitude for the respective band. Specifically, for the VI color, the reddening-free magnitude for the J band should be W_J^{VI} , not W_V^{VI} . This implies that ΔM_J must be correlated with ΔW_J^{VI} . The slopes of the regression lines for the four families of Wesenheit functions—based on BI, VI, IH, and JK—are summarized in Table 2.1.

Band	ρ_{λ}^{BI} (error)	ρ_{λ}^{VI} (error)	ρ_{λ}^{IH} (error)	ρ_{λ}^{JK} (error)
B	0.797 (± 0.088)	0.819 (± 0.075)	0.791 (± 0.094)	0.619 (± 0.111)
V	0.818 (± 0.066)	0.847 (± 0.060)	0.882 (± 0.079)	0.778 (± 0.095)
I	0.906 (± 0.041)	0.907 (± 0.036)	1.022 (± 0.049)	0.993 (± 0.060)
J	0.979 (± 0.022)	0.976 (± 0.019)	1.013 (± 0.023)	1.037 (± 0.028)
H	0.986 (± 0.014)	0.984 (± 0.013)	1.007 (± 0.015)	1.023 (± 0.018)
K	0.994 (± 0.009)	0.992 (± 0.008)	1.007 (± 0.010)	1.015 (± 0.012)

Table 2.1: Slopes of residual correlation $\Delta W_{\lambda}^{12} - \Delta M_{\lambda}$ for BI, VI, IH and JK. Note the systematic decrease in errors as the slope ρ approaching to 1 with increasing wavelength.

2.3 Decoupling $\delta\mu - \delta E_{BV}$

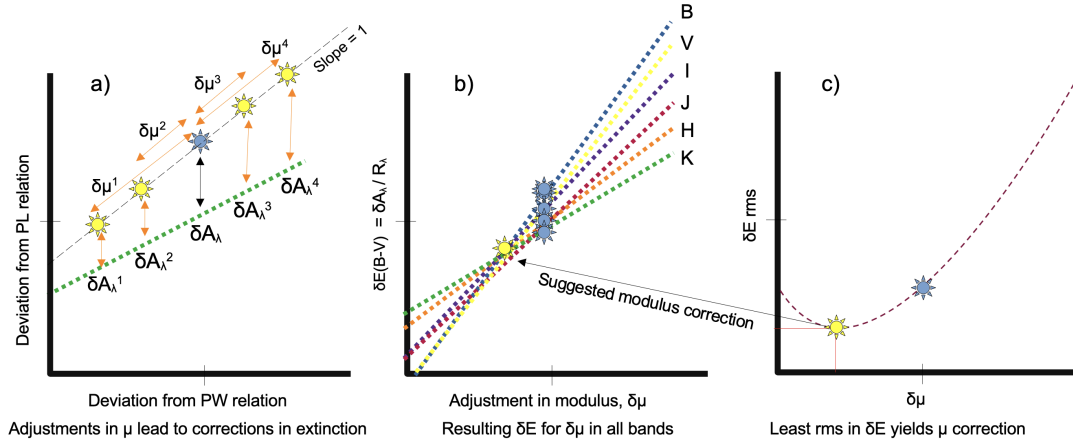


Figure 2.10: The key concept behind error degeneracy

The core process of the calibration is encapsulated in this step. Up until this point, from the Galactic Cepheid dataset, I have derived multiband Leavitt laws and their corresponding Wesenheit Leavitt laws. The correlation of their residuals leads to the results summarized in Table 2.1, which will be used for decoupling the errors δE_{BV} and $\delta\mu$ for each star in the dataset. Given the high sensitivity of reddening at shorter wavelengths, to estimate a realistic correction for reddening error (δE_{BV}), averaging result from the first four bands — B, V, I, and J would be enough.

Since the vertical deviation from the regression line in $\Delta W_{\lambda}^{12} - \Delta M_{\lambda}$ suggested the extinction error, mathematically, it can be formulated as:

$$\delta A_{\lambda}^0 = \Delta M_{\lambda} - \rho_{\lambda}^{12} \Delta W_{\lambda}^{12} - \sigma_{\lambda}^{12} \quad (2.5)$$

The superscript '0' on A indicates that the distance error trial is not considered yet. ρ_{λ}^{12}

represents the slope of the ΔW_λ^{12} vs. ΔM_λ plot. Since residuals are dispersed around the origin, intercept of the regression lines σ_λ approach to zero and can be neglected.

To convert the extinction error into reddening error, dividing by R_λ^{BV}

$$\delta E^0(B - V)_\lambda = \frac{\delta A_\lambda^0}{R_\lambda^{BV}} \quad (2.6)$$

By making trials for the error in the modulus, $\delta\mu^i$, the corresponding reddening error, δA_λ^i , is calculated as follows:

$$\begin{aligned} \delta A_\lambda(\delta\mu^i) &= (\Delta M_\lambda + \delta\mu^i) - \rho_\lambda^{12}(\Delta W_\lambda^{12} + \delta\mu^i) \\ &= (\Delta M_\lambda - \rho_\lambda^{12}\Delta W_\lambda^{12}) + \delta\mu^i - \rho_\lambda^{12}\delta\mu^i \\ &= \delta A_\lambda^0 + \delta\mu^i - \rho_\lambda^{12}\delta\mu^i \end{aligned}$$

Therefore, for given modulus correction trail, extinction in any band will be calculated as follow:

$$\delta A_\lambda^i = \delta A_\lambda^0 + \delta\mu^i(1 - \rho_\lambda^{12}) \quad (2.7)$$

$$\delta E_\lambda^i(B - V) = \frac{\delta A_\lambda^i}{R_\lambda^{BV}} \quad (2.8)$$

$$(2.9)$$

Considering ideal case where extinction law $\frac{A_\lambda}{A_V}$ and reddening ratio R_λ are precisely known for the line-of-sight of each Cepheid of the dataset. Let i number of trails made for modulus correction $\delta\mu$ which correspond to six variants of reddening corrections (BVIJHK) for each weseneit-color (BI, VI, IH and JK). Each Cepheid yields an error matrix of 100 rows ($\delta\mu^i$) and 6 x 4 columns for expected $\delta E_\lambda^i(B - V)$.

Let, k^{th} trail be the correct one, i.e. reddening corrections δE_λ suggested by each band must be agreeing with remaining ones., $\delta\mu^k$ must yield the same reddening correction δE^k for all the bands, impling their variance is zero.

$$\delta E_{BV}(\delta \mu^k) = \delta E_B^k = \delta E_V^k = \delta E_I^k = \delta E_J^k = \delta E_H^k = \delta E_K^k = \langle \delta E_\lambda^k \rangle_\lambda \quad (2.10)$$

For dataset, it the variance across the bands being the minimum for the best solution, when .

$$k = \min(\text{var}(\delta E_\lambda^i(B - V))_\lambda)_i \quad (2.11)$$

2.4 Calibrated Leavitt Law

These corrections will be adjusted with the original data as follow.

$$M_\lambda^* = M_\lambda^0 + \delta A_\lambda^* + \delta \mu^*$$

Gaia parallax based distances are independent of reddening error as the method is fundamentally geometrical in nature. Although, Cepheid being one of the brightest target in the sky, leading to saturation of photons on the CCD causing a strong bias for longer period Cepheids. This makes parallax of Cepheid slightly uncertain as compare to dimmer stars. I have implemented the calibration algorithm on the same set of stars and determine calibrated distance for individual Cepheid. If the algorithm works perfectly, calibrated distance must agree with IRSB based calibrated distance.

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9_plots/6_rms/95_1_star_SBI_g.pdf

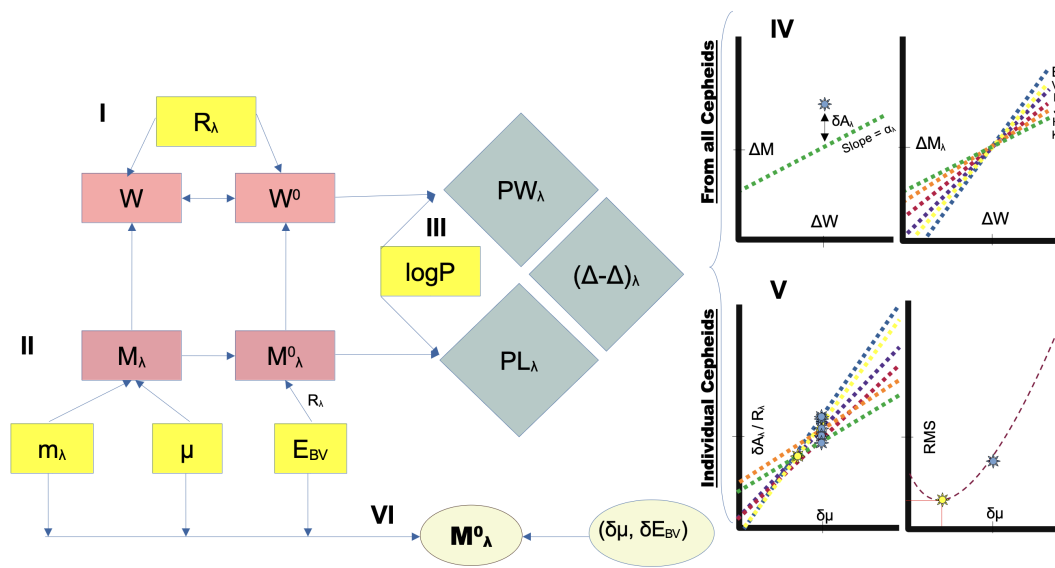


Figure 2.12: The Algorithm

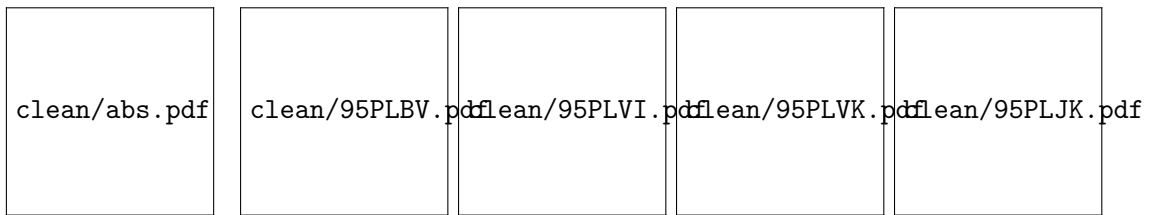


Figure 2.13: BVIJHK PL and PW relations with their Pearson correlation coefficient (r-value).

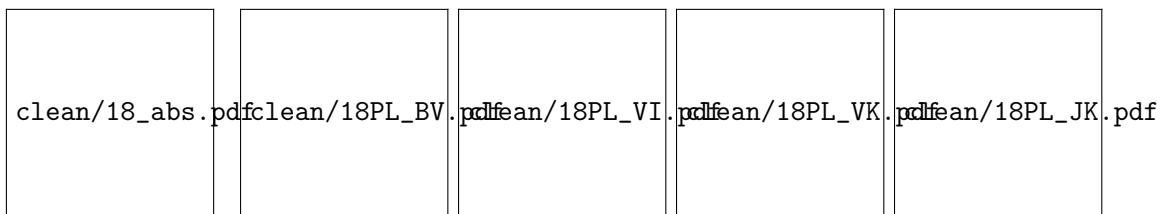


Figure 2.14: BVIJHK PL and PW relations with their Pearson correlation coefficient (r-value).

Part III

Result Implications

3 Discussion

3.1 Validation of Result : Cluster Cepheid

In the dataset, 17 Cepheids are identified as members of clusters with statistical parallaxes measured by cruz2023. A comparison between the calibrated distances of these Cepheids shows a good agreement in reddening value with $\sigma = 0.049$.

For distance modulus, nine of 17 cluster Cepheid deviated by 0.1 mag only, but five Cepheids (EV Sct, QZ Nor, SV Vul, Y Sct and X Lac) have deviation larger than 0.4 mag.

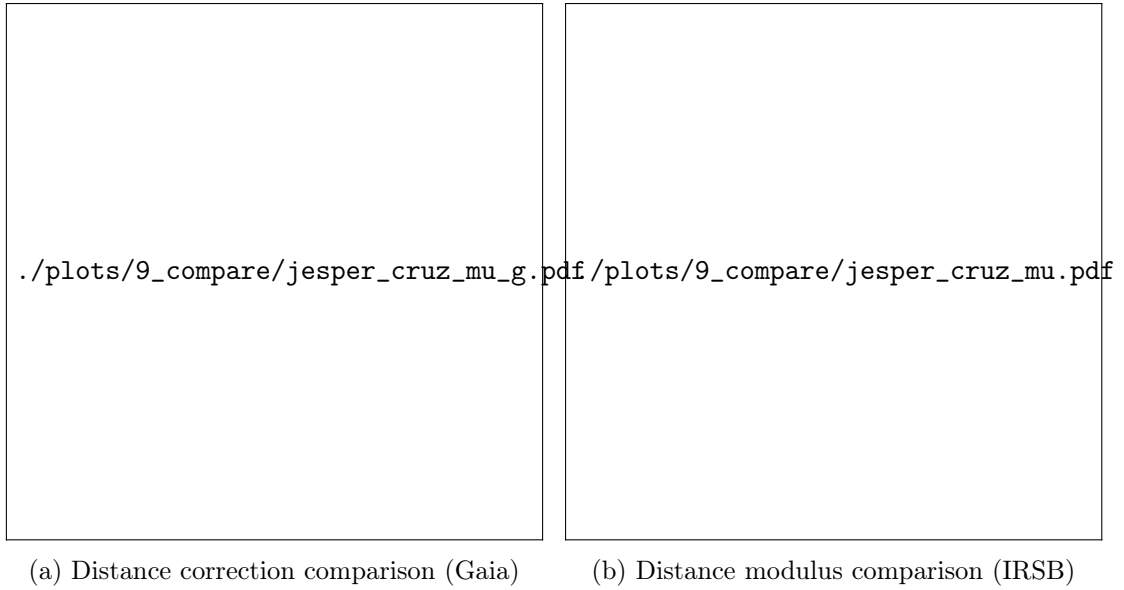


Figure 3.1: Deviation in corrected distance with 17 Cluster Cepheids (Cruz, 2023)

metallicity is wrong then color is too

3.2 Reddening Comparison

(?)

3.3 Distance Comparison

3.4 Luminosity Comparison

Effect of Metallicity

While metallicity can affect the zero-point and slope of the period–luminosity relation, it is not explicitly considered in this thesis. However, the framework developed here is readily extendable to incorporate metallicity corrections in future studies, enabling a more comprehensive calibration of the Leavitt Law across different galactic environments.

Limitation of the method

The method primarily does not calibrate the Leavitt Law itself, rather correct distance and reddening of individual Cepheids according to the line of linear regression for given set of Cepheid. Therefore, for different dataset, correction in distance and reddening could be slightly different due to the selection bias. To minimize such effect, I had omitted such Cepheids from the analysis which cause significant bias to regression line of residual correlation.

4 Conclusion

4.1 Result

4.2 Application Of The Result

LMC and SMC Distances

To evaluate the effectiveness of the calibration, distances to the Large and Small Magellanic Clouds (LMC and SMC) are derived using 24 versions of the calibrated Leavitt Law, covering four photometric bands (VIJK) and six color-based Wesenheit functions. This approach allows the construction of a distance matrix that highlights the relative performance of different color indices in producing consistent distance estimates. The results demonstrate the sensitivity of the PL calibration to the choice of photometric band and Wesenheit function, providing guidance for selecting the most robust combination for extragalactic distance determinations.

4.2.1 Understanding the Universe: 1990s

A major milestone in the extragalactic distance ladder was achieved in 1998 with the incorporation of Type Ia supernovae as standardizable candles. The development of robust empirical methods to measure distances to galaxies hosting these supernovae, calibrated using the Cepheid period–luminosity relation, extended reliable distance measurements from scales of ~ 100 million light-years to several billion light-years. This dramatic extension of the cosmic distance scale led to one of the most surprising discoveries in modern cosmology: the expansion of the Universe is accelerating.

This acceleration implies the presence of a component with negative effective pressure that drives the expansion of spacetime. This unknown component was termed *dark energy*. Despite extensive observational and theoretical efforts, the physical nature of dark energy remains unknown. Its discovery has underscored the importance of understanding not only the geometry of spacetime but also the detailed composition of

the Universe. A simplified description of the Universe can be framed in terms of its total mass–energy content and the geometry it induces, where energy appears in various forms such as mass, radiation, and thermal energy.

In the standard cosmological framework, the Universe is composed primarily of three components: baryonic matter, dark matter, and dark energy. Ordinary, or *baryonic*, matter consists of protons, neutrons, and electrons and accounts for only about 5% of the total energy density of the Universe. This component forms the stars, galaxies, and large-scale structures directly observable through electromagnetic radiation.

Dark matter constitutes approximately 27% of the total energy content of the Universe. It is hypothesized to possess fundamentally different properties from baryonic matter, as it

neither emits nor absorbs electromagnetic radiation, rendering it invisible to direct observation. Its existence is inferred through its gravitational influence on visible matter and radiation. Strong observational evidence for dark matter arises from gravitational lensing, galaxy rotation curves, and the fine-scale anisotropies observed in the cosmic microwave background (CMB). Importantly, the interpretation of these phenomena depends critically on accurate distance measurements; for example, lensing geometries and dynamical mass estimates are highly sensitive to the assumed distances of the relevant systems. Despite its well-established gravitational effects, the physical nature of dark matter remains one of the most significant open questions in contemporary astrophysics.

Dark energy dominates the cosmic energy budget, contributing approximately 68% of the total energy density of the Universe. In contrast, all other components—including radiation, neutrinos, and possible exotic particles—together contribute less than 1%. Determining the relative proportions of these components relies heavily on precise distance measurements across cosmic scales, as such measurements underpin estimates of luminosity, redshift, and volume, all of which are essential for reconstructing the Universe’s composition and its evolutionary history.

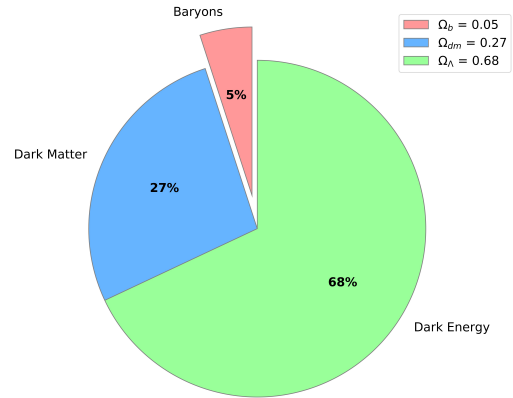


Figure 4.1: The fractional composition of the Universe consists of dark energy (68 %), dark matter (27 %), and baryonic matter (5 %) is depicted, with the contributions from radiation and other relativistic particles neglected.

4.2.1.1 Λ CDM Cosmological Model

Observations of the distant Universe allow us to probe earlier cosmic epochs, as light requires finite time to traverse space. This fundamental property enables direct investigation of the Universe's evolution. One approach to characterizing cosmic expansion involves measuring distances to the most distant observable objects, effectively constraining the size of the observable Universe and the cosmological horizon.

By combining insights from high-energy particle physics with general relativity, cosmologists have been able to model the evolution of the Universe back to fractions of a second after the Big Bang. Integrating these theoretical frameworks with a broad range of observational data has led to the formulation of the standard cosmological model, known as the Λ CDM (Lambda Cold Dark Matter) model. This model is characterized by a small set of parameters, including the densities of dark energy, dark matter, and baryonic matter; the Hubble constant (H_0); the scalar spectral index describing primordial density fluctuations; and the curvature parameter that determines the geometry of spacetime. Many of these parameters are tightly constrained by measurements of the cosmic microwave background and are independently supported by large-scale structure surveys, baryon acoustic oscillations, and supernova observations.

The full-sky temperature anisotropy map shown on the introductory page is a snapshot of the early Universe obtained by the *Planck* satellite, a mission of the European Space Agency operational between 2009 and 2013. This map depicts the cosmic microwave background radiation, the relic thermal emission from the epoch of recombination. The small temperature fluctuations observed across the sky correspond to primordial density perturbations in the early Universe. Over billions of years, these perturbations grew through gravitational instability, ultimately giving rise to galaxies, clusters, and the large-scale structure observed today.

The CMB represents the earliest observable relic of the Universe, providing a view of conditions approximately 380,000 years after the Big Bang. Its precise measurement enables robust constraints on fundamental cosmological parameters, including the age, composition, and expansion rate of the Universe. Among the most significant results derived from the *Planck* mission is a high-precision determination of the Hubble constant, H_0 , which plays a central role in understanding both the past evolution and the future fate of the Universe.

4.2.2 Cosmic Distance Ladder and the Hubble Constant: The Twenty-First Century

Distance measurements across cosmic scales rely on a hierarchical set of interconnected methods collectively known as the *cosmic distance ladder*, in which each rung is calibrated using more local and direct techniques. This framework allows distances to be extended progressively to cosmological scales, where they can be compared with early-Universe inferences from the cosmic microwave background (CMB).

A major refinement of the ladder was achieved by Riess and collaborators using Type Ia supernovae (SNIa) as distance indicators. These thermonuclear explosions of white dwarfs exhibit a well-defined correlation between peak luminosity and light-curve shape, enabling their use as standardizable candles. By calibrating SNIa luminosities with Cepheid variable stars via the Leavitt Law, distances were extended from the Cepheid-accessible local volume of ~ 30 Mpc ($\sim 10^8$ ly) to beyond 1000 Mpc ($\sim 10^{10}$ ly) (RF98).

Figure 4.2 illustrates the cosmic distance ladder, beginning with geometric parallax, progressing through Cepheids, and culminating in SNIa measurements at high redshift. Applying this calibrated framework to the Hubble–Lemaître relation yields

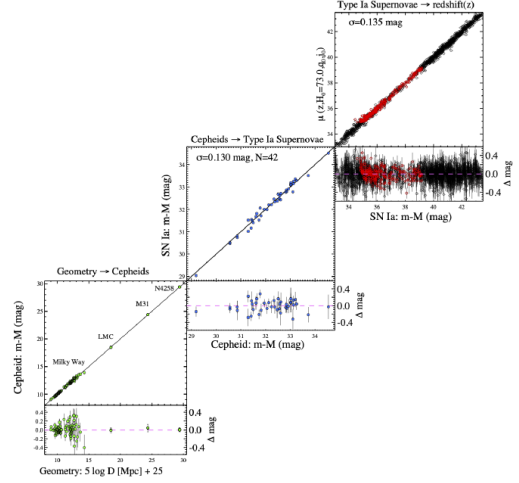


Figure 4.2: An instance of Cosmic Distance Ladder calibrating Leavitt Law with parallax, SNIa with Leavitt Law and Hubble Law with SNIa based distances. (RYM⁺22)

$$H_0 = 73.30 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

which differs by $\sim 5\sigma$ from the value inferred by *Planck* CMB observations under Λ CDM assumptions (RYM⁺22; AAA20). This persistent discrepancy, known as the *Hubble tension*, may reflect unaccounted systematic uncertainties or signal new physics beyond the standard cosmological model, and remains a central challenge in contemporary cosmology.

4.2.3 Instruments of future

The James Webb Space Telescope (JWST), launched in 2021, complements Gaia by extending observations into the infrared, where Cepheids are less affected by dust

extinction and metallicity effects. JWST's sensitivity allows high-precision photometry of Cepheid variables in nearby and distant galaxies, further refining their period–luminosity relation and extending the reach of the cosmic distance ladder deeper into the Hubble flow.

In addition to these missions, major surveys and observatories such as SDSS, Planck, and 2dFGRS have provided critical data for cross-checking and anchoring distance measurements. Future facilities—including the Vera C. Rubin Observatory, DESI, 4MOST, Euclid, SKA, and LISA—promise to deliver even more precise observations, offering independent calibration pathways for standard candles and the local expansion rate.

Beyond photometry and astrometry, the advent of multimessenger astronomy—through the detection of cosmic rays, neutrinos, and gravitational waves—introduces alternative distance indicators, such as standard sirens, which can independently validate Cepheid- and SNIa-based distances.

Collectively, these technological and observational advances mark the golden era of modern astronomy. By providing highly accurate distances, they enable precise calibration of the Leavitt Law, strengthening the foundation of the cosmic distance ladder and advancing our ability to determine the Hubble constant with reduced systematic uncertainties.

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